



TransConv: convolution-infused transformer for protein secondary structure prediction

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Abstract

Context Protein secondary structure prediction is essential for understanding protein function and characteristics and can also facilitate drug discovery. Traditional methods for experimentally determining protein structures are both time-consuming and costly. Computational biology offers a viable alternative by predicting protein structures from their sequences. Protein secondary structure is defined by the local spatial arrangement of the protein backbone, resulting from hydrogen bonds between amino acids.

Methods In this study, we introduce TransConv, a model that combines transformer models with convolutional blocks to predict protein secondary structures from amino acid sequences. Transformer models are effective at capturing long-range dependencies through self-attention mechanisms. We integrate convolutional blocks into the transformer architecture to improve the detection of important local features. This hybrid model captures both long-range interactions and local features, leading to more accurate predictions of protein secondary structures, thus offering an efficient solution for this critical task. The experimental outcomes on the benchmark datasets depict the superiority of the proposed approach over the state-of-the-art (SOTA) models in the literature.

Keywords Transformer · Convolutional features · Protein · Secondary structure

Introduction

Proteins are fundamental to life, performing a vast array of essential biological functions [1]. These macromolecules are crucial in almost every cellular process, including catalyzing metabolic reactions, replicating DNA, responding to stimuli, and transporting molecules [2]. The functionality of proteins is closely linked to their structure, which can be understood at various levels [3].

The primary structure of a protein is its linear sequence of amino acids, connected by peptide bonds to form a polypeptide chain [4]. In the human body, these polypeptide chains undergo spatial coiling, bending, and folding due to interactions among the side chains of the amino acids, leading

to a unique three-dimensional conformation known as the tertiary structure [5]. This complex folding is driven by various chemical interactions, including hydrogen bonds, ionic bonds, van der Waals forces, and hydrophobic interactions, all of which contribute to the protein's stability and functionality.

Understanding protein structure is crucial for comprehending protein function because a protein's shape largely determines how it interacts with other molecules. Experimental methods such as X-ray crystallography and nuclear magnetic resonance (NMR) spectroscopy can determine protein structures with high accuracy, but these techniques are both costly and time-consuming, requiring substantial resources and expertise. To mitigate these limitations, machine learning models can be utilized to predict protein structures from amino acid sequences. However, predicting the tertiary structure from protein sequences is one of the most challenging tasks in computational biology due to the vast conformational space a protein can occupy and the complexity of the forces that stabilize its final structure [6].

The initial step in addressing this complex task is predicting the protein's secondary structure. This intermediate level

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of protein structure provides valuable insights into a protein's activity, relationships, and functions. The secondary structure is defined by the local spatial arrangement of the protein backbone, including common motifs such as alpha helices and beta sheets, which are primarily stabilized by hydrogen bonds between the carbonyl oxygen and amide hydrogen atoms in the protein backbone [7]. Understanding these motifs can greatly enhance our knowledge of a protein's overall architecture and functional sites.

To effectively tackle this challenge, a model must capture the intricate dependencies between various amino acids within the protein sequence. This involves recognizing patterns and long-range interactions that influence folding. Advanced machine learning techniques, such as deep learning and neural networks, have shown promise in modeling these complex dependencies. By training on large datasets of known protein structures, these models can learn to predict secondary and tertiary structures with increasing accuracy. Consequently, computational approaches hold the potential to accelerate biological research and drug discovery by providing faster and more cost-effective means of elucidating protein structures.

The transformer model, initially developed for natural language processing (NLP) [8], has proven to be versatile across various domains. In the context of protein structure prediction, where understanding relationships among amino acids is critical, the transformer's attention mechanism offers significant advantages. To enhance the model's ability to discern both local and global features within protein embeddings, this work proposes a model, TransConv, which incorporates convolutional features into the transformer encoder. This combination aims to leverage the transformer's proficiency in capturing long-range dependencies and the convolutional model's effectiveness in highlighting prominent local features within the protein sequence.

Addressing the challenges posed by limited dataset availability in protein structure prediction, traditional methods often incorporate evolutionary information from broader databases. Multiple sequence alignment (MSA) and position-specific scoring matrix (PSSM) are derived from homologous proteins, augmenting protein sequence data for more accurate predictions [9, 10]. However, a significant limitation arises as a majority of proteins lack substantial homologous sequences [11]. To overcome this hurdle, we utilize pre-trained language models fine-tuned on vast unlabeled datasets to generate protein embeddings without relying on evolutionary information. Following the success of previous studies [12, 13], we employ the ProtT5-XL-U50 pre-trained protein language model [12] to generate protein embeddings, derived from the T5 transformer model [14].

The primary contribution of this work is the integration of convolution operations with the attention mechanism of the transformer encoder model to efficiently identify the

relationships between amino acids in a protein sequence and predict secondary structures from the given sequence. Furthermore, the experiments are carried out on benchmark datasets, such as ProteinNet [15], NetSurfP-2.0 [16], and NEW364. The experimental results of the proposed approach are extensively compared with the state-of-the-art (SOTA) models in the existing literature. The experimental outcomes and its comparative analysis clearly show the superiority of the proposed TransConv approach over the SOTAs.

The remaining sections of the paper are organized as follows: the “[Background](#)” section presents some preliminaries for this work and the “[Related works](#)” section provides a detailed overview of the related works. In the “[Proposed methodology](#)” section, the proposed methodology is discussed in detail. The “[Experimental setups](#)” section contains the details about the experiments conducted, including details about the dataset and the parameters used during training. In the “[Results and discussions](#)” section, the results of the experiments conducted are presented in detail. The “[Conclusion](#)” section wraps up the work with some concluding remarks and a discussion of future research directions.

Background

Transformer

In 2017, Vaswani et al. introduced the transformer model [8], revolutionizing the field of natural language processing (NLP) and establishing a new standard for sequence-to-sequence tasks in this domain. This innovative architecture is particularly adept at converting a sequence of elements, such as words in a sentence, from one language to another—a process widely known as language translation.

However, the versatility of the transformer model extends far beyond language translation, finding applications in various sequence-based tasks [17]. One such application is in predicting protein secondary structures, where the transformer model is used to transform a sequence of amino acids into a corresponding sequence of secondary structure assignments. This adaptation is noteworthy because the principles that enable the transformer to comprehend and generate human language are equally effective in deciphering the “language” of proteins, where amino acids function similarly to words. Our approach emphasizes the use of an encoder-only architecture, diverging from the traditional encoder-decoder configuration prevalent in NLP tasks.

As depicted in Fig. 1, the transformer encoder model begins by processing input embeddings of a specified dimension, denoted as d_{model} .

To maintain the sequence order of the input elements, the model employs positional encodings, which are added to the input embeddings. The encoder is composed of N

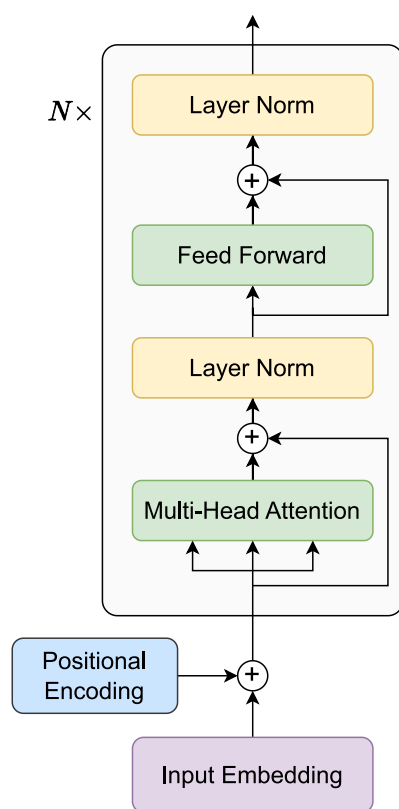


Fig. 1 Diagram of transformer encoder model

identical layers, each containing two distinct sublayers. The first sublayer is a multi-head self-attention mechanism, while the second is a fully connected feed-forward network. A key architectural feature is the use of residual connections around each sublayer, followed by layer normalization [18], formalized as $Layer Normalization(x + Sublayer(x))$. These residual connections facilitate the smooth flow of information through the network by bypassing non-linear activation functions, ensuring efficient and effective processing of the input sequence.

Protein secondary structure

Proteins, fundamental to all living organisms, are composed of chains of amino acids. Each amino acid is represented by a unique one-letter code from the set {"A," "C," "D," "E," "F," "G," "H," "I," "K," "L," "M," "N," "P," "Q," "R," "S," "T," "V," "W," "Y"}. For example, "A" stands for Alanine, "C" for Cysteine, "D" for Aspartic Acid, among others.

The linear sequence of these amino acids constitutes the protein's primary structure [19], essentially a one-dimensional (1D) arrangement. Within the complex environment of the human body, these proteins twist, bend, and fold into specific three-dimensional (3D) shapes, referred to as their tertiary structure [5]. Predicting this 3D structure

directly from the primary structure is a significant and ongoing challenge in current scientific research. Hence, our focus is on predicting the secondary structure of proteins, which serves as an intermediate step linking the primary structure to the tertiary structure.

The secondary structure [20] provides crucial insights into the local characteristics of proteins and is derived from the tertiary structure using algorithms such as DSSP (define secondary structure of proteins). The DSSP algorithm carefully analyzes hydrogen bond formations between amino acid pairs based on the protein's 3D coordinates. It then assigns one of eight distinct secondary structure types (DSSP8): G, H, I, T, B, E, S, and C (or L), as illustrated in Table 1.

Furthermore, an additional blank symbol within these eight categories is used to denote unknown or padding elements in the sequence. Thus, the output vocabulary of our model comprises a total of nine characters, reflecting the diverse array of protein secondary structure classifications.

Related works

Over the past few decades, significant advancements have been made in the field of protein structure prediction. Modern deep learning techniques, with enhanced capabilities for capturing long-range interactions, have emerged as the forefront of protein secondary structure prediction methods [21].

Hansen et al. [22] employed an ensemble approach using LSTM-BRNN and ResNet models, supplemented by predicted residue-residue contact maps, to forecast both 3-state and 8-state secondary structures. This approach has shown promising results in improving prediction accuracy.

Heinzinger et al. [23] introduced two deep learning models: DeepProtVec and DeepSeqVec. DeepProtVec utilizes a fundamental deep learning algorithm based on convolution with a window size of 7, trained on prediction tasks using ProtVec, a popular embedding tool for protein sequences derived from the Word2vec methodology. On the other hand, DeepSeqVec shares the same architecture as DeepProtVec

Table 1 Types of DSSP8 assignments

Description	DSSP8
3-Helix	G
α -Helix	H
5-Helix (pi helix)	I
Residue in isolated β -bridge	B
Extended strand participates in β ladder	E
Bend	S
Hydrogen bonded turn	T
Coil/Loop	C/L

but employs SeqVec for input encoding. SeqVec (Sequence-to-Vector) is introduced as a novel method for representing protein sequences as continuous vectors (embeddings) using the ELMo language model.

Klausen et al. developed NetSurfP-2.0 [16], a sequence-based tool that combines convolutional and long short-term memory neural networks trained on solved protein structures. NetSurfP-2.0 uses a single integrated model to predict solvent accessibility, secondary structure, structural disorder, and backbone dihedral angles for each residue in the input sequences.

In another study, Wang et al. [24] proposed the Deep-CNF model, which extends conditional neural fields (CNF) with deep learning techniques, demonstrating enhanced prediction performance. Furthermore, Xu et al. [25] introduced a transformer-based model with a modified architecture designed to theoretically capture interactions between residues at arbitrary distances. They also utilized an ensemble of models to achieve improved final results. Moreover, Rives et al. [26] presented ESM-1b, a transformer-based model with 33 layers, 650 million parameters, and embeddings of

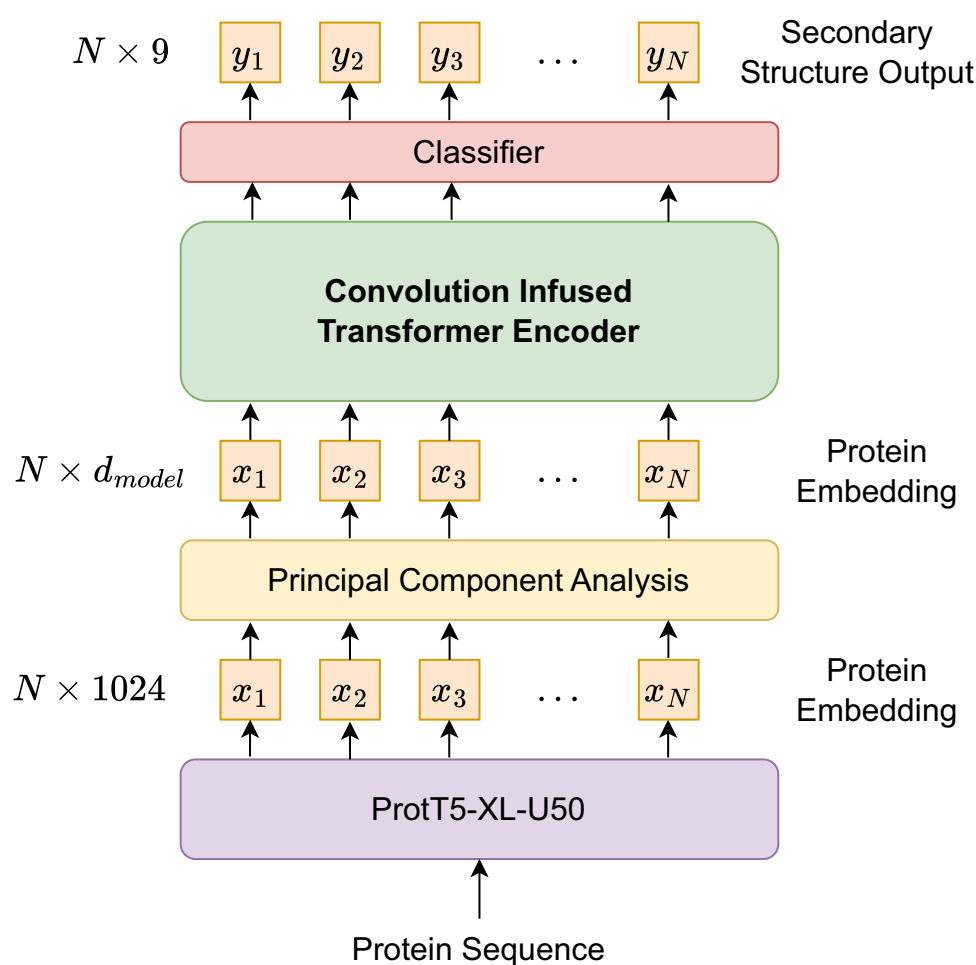
size 1280. ESM-1b serves as a protein language model leveraging the transformer architecture.

While transformer models have shown potential in predicting protein secondary structure, there remains an opportunity to explore the integration of attention mechanisms and convolution operations within a transformer encoder to further enhance prediction accuracy.

Proposed methodology

In the proposed enhanced transformer encoder model, the primary structure sequence of a protein is used as the input. Although protein sequences can vary in length, the transformer model effectively handles this variability. The output generated by the transformer encoder model will match the input sequence length. Predicting secondary structures solely from the protein sequence is inherently challenging, and traditional methods often incorporate evolutionary data from extensive datasets. Techniques such as multiple sequence alignment (MSA) and position-specific scoring

Fig. 2 Schematic overview of the proposed model



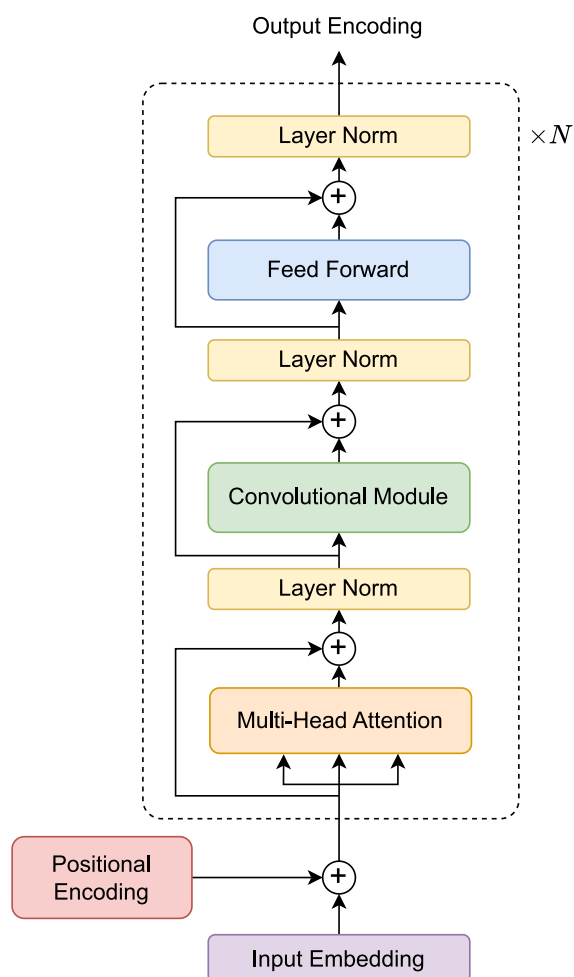


Fig. 3 Architectural diagram of the proposed convolution-infused transformer encoder model

matrix (PSSM), derived from homologous proteins, enhance prediction accuracy by enriching the protein sequence data [9, 10].

Nevertheless, a significant constraint is that a large portion of proteins do not have sufficient homologous sequences [11]. To overcome this limitation, this study integrates pre-trained language models. We utilize these models, which have been fine-tuned on extensive unlabeled datasets, to generate protein embeddings without depending on evolutionary information. Building on the success of previous research [12, 13], we employ the ProtT5-XL-U50 pre-trained protein language model [12] to produce these embeddings. This approach aims to improve the accuracy of secondary

structure prediction in proteins that lack significant homologous sequences.

In this research, we begin by inputting the protein sequence into the pretrained Prot-XL-U50 model, which produces protein embeddings from the given amino acid sequence, as shown in Fig. 2.

These embeddings have a dimensionality of 1024. The embeddings are intended for input into a subsequent transformer model. However, using a 1024-dimensional embedding for each amino acid may pose significant computational challenges. To mitigate this issue, we employ principal component analysis (PCA) to reduce the dimensionality of the embeddings. This reduction makes it feasible to use a smaller and more efficient transformer model in the later stages of the protein structure prediction process.

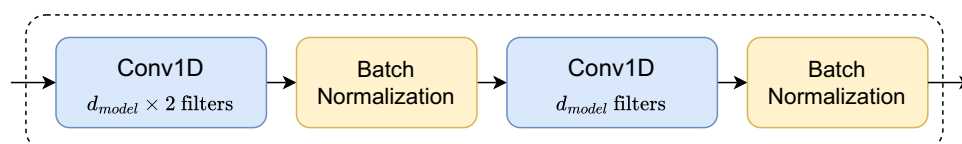
With the reduced-dimensional embeddings, we introduce a modified transformer model. This model begins by integrating learnable positional encoding (LPE) to retain the sequential order of the input embeddings. The objective is to enable the transformer to effectively capture the relative positions of amino acids within the sequence. By incorporating LPE, the model can effectively encode positional information alongside the contextual information embedded in the protein embeddings, ensuring that the transformer accurately identifies the spatial arrangement of amino acids within the sequence.

The proposed transformer encoder model primarily consists of three sublayers: a multihead self-attention (MHSA) module for capturing long-range dependencies within the input sequence, followed by a convolutional module to capture local dependencies among the sequences, and finally, a fully connected feed-forward neural network. The proposed infused transformer encoder model is depicted in Fig. 3.

Each of these sublayers is connected by residual connections and layer normalization operations. The convolutional module comprises two 1-D convolution operations and batch normalization, as presented in Fig. 4.

The modified encoder then outputs an encoded sequence that contains information about both local and long-range dependencies within the input protein sequence. This encoded output is subsequently passed to a multilayer perceptron (MLP) classifier, which classifies each position of the sequence into one of the possible secondary structure assignments. If the input size of the proposed model is $N \times d_{\text{model}}$, where N is the length of input protein sequence and d_{model} is the size of protein embeddings obtained after PCA, then the model yields an output size of $N \times 9$. This accounts for the

Fig. 4 Detailed diagram of the convolution module



8 possible DSSP8 assignments and an additional symbol for the padding token.

By integrating these steps, we establish a comprehensive framework for predicting protein secondary structure. This approach leverages pretrained language models, dimensionality reduction, and a modified transformer architecture with convolutional enhancements, collectively aiming to optimize predictive accuracy and capture intricate relationships within protein sequences.

Experimental setups

Dataset description

The proposed model is trained on two distinct datasets to enable an effective comparison with baseline models. The first dataset used is ProteinNet [15], a benchmark dataset for protein structures derived from PDB structures available as of 2016. ProteinNet provides data at various sequence identity cutoffs; for our study, we selected a dataset with a 95% cutoff, aligning with previous research. This training dataset consists of 39,120 sequences, but due to incomplete secondary structure information in ProteinNet, only 38,000 sequences could be annotated using the DSSP program. For validation, we employed 100 proteins, identical to those utilized in prior studies.

The trained model was then evaluated on several test datasets, including SPOT-2016, SPOT-2016-HQ, SPOT-2018, SPOT-2018-HQ, and TEST-2018. These test sets, like the training dataset, comprise protein structures released up to 2016. The SPOT-2016 dataset includes proteins released between 2016 and 2020, excluding those with an e-value cutoff less than 0.1. SPOT-2018 and SPOT-2018-HQ are subsets of SPOT-2016, featuring proteins released post-2018 and meeting specific resolution criteria. The TEST-2018 dataset contains high-resolution proteins released in 2018, filtered to maintain a 25% sequence identity threshold relative to pre-2018 proteins.

The second dataset utilized is NetSurfP-2.0 [16], which offers 10,792 data samples in both 3-state and 8-state DSSP

formats. This dataset, downloaded in CSV format, was used for model validation with 646 protein samples, including those from CASP12, CB513, and TS115. These independent datasets have been used in previous research. Proteins with sequence similarities greater than 25% to any protein in these validation sets were excluded from the training set, though redundancy among the test datasets was not addressed.

To complement the limitations of the aforementioned test sets, we introduced the NEW364 dataset. This dataset includes proteins selected from the PDB with a resolution of 2.5 Å or better and a minimum length of 20 amino acids. Proteins published after 2019 were included in NEW364. Tools such as MMSeqs and PISCES were employed to eliminate proteins with more than 20% similarity to either the training data or within the dataset itself, ensuring a diverse and independent evaluation set for the model trained with NetSurfP-2.0.

Training details

The proposed TransConv model employs an input embedding size of 128. It features 10 encoder layers, each comprising 8 attention heads. Additionally, each encoder layer integrates a convolutional module, depicted in Fig. 4, which primarily includes two 1D convolution layers with kernel sizes of 7 and 5, respectively. The feedforward layer within the encoder possesses a dimensionality of 512.

In this research, the Adam optimizer, characterized by a variable learning rate as illustrated in Fig. 5, is used to train the TransConv model. The learning rate is managed by a scheduler similar to the one presented by Vaswani et al. [8], which incorporates a warmup period spanning 5 epochs. This warmup phase allows the learning rate to increase gradually during the initial epochs, contributing to the stabilization of the training process. The momentum parameters of the Adam optimizer, denoted as β_1 and β_2 , are set at 0.9 and 0.999, respectively.

Throughout the experimental phase, a constant batch size of 8 is employed, and the training is conducted for a maximum of 50 epochs. Figures 6 and 7 display the graphs of training and validation accuracies over the course of

Fig. 5 Learning rate variation through epochs

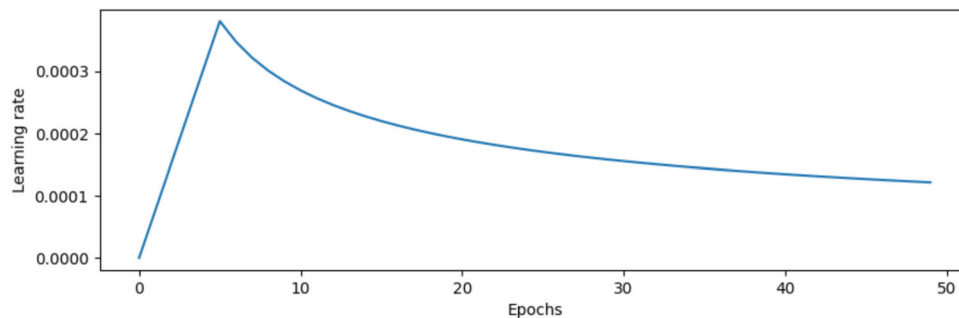
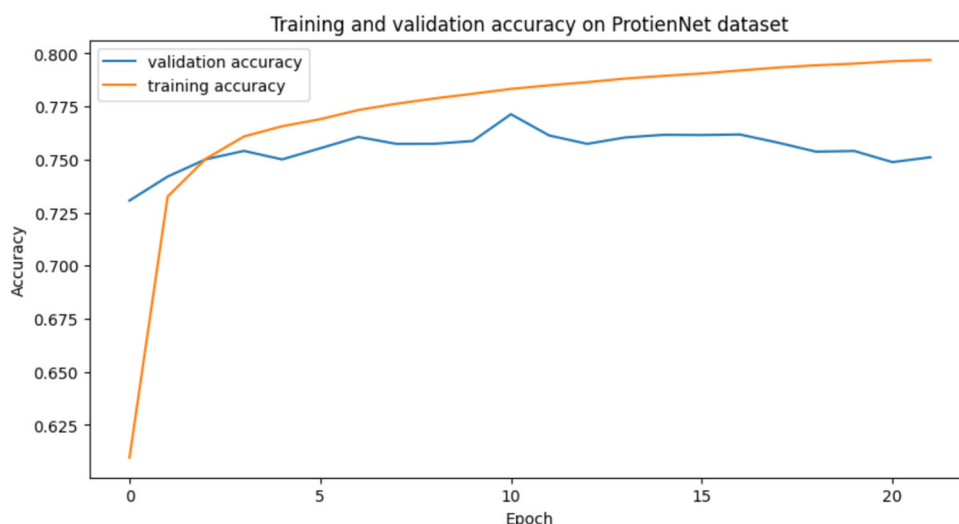


Fig. 6 Training and validation accuracy graph for training on ProteinNet dataset



the training process for the ProteinNet and NetSurfP–2.0 datasets, respectively. It is noteworthy that, in our experiments with the ProteinNet dataset, we observe a decrease in validation set accuracy from the 10th epoch onward, indicative of potential overfitting. This behavior is commonly observed in deep learning models, especially when the model complexity exceeds the available training data or when training continues beyond the optimal point. To address this, we have used dropout layers in multiple places in our model, mainly after the MultiHeadAttention module, convolutional module and feedforward layer inside the transformer encoder. The dropout value is set to quite a high value of 0.5 to mitigate overfitting. Our findings suggest that while the model initially learns the underlying patterns in the training data, it begins to memorize the training examples rather than generalizing to unseen data as training progresses. Thus, the observed decline in validation accuracy after the 10th epoch is expected and emphasizes the importance of

utilizing strategies such as cross-validation and regularization in future iterations of model training.

Experimental setup

The experiments detailed in this study were performed using Python version 3.10.11. The following libraries were employed: torch version 2.0.1, numpy version 1.24.4, scikit-learn version 1.3.0, and pandas version 2.0.3. For executing the pretrained ProtT5-XL-U50 model, we utilized the Hugging Face Transformers module, version 4.32. All computational tasks were carried out on a Dell Precision 7820 workstation. This workstation is equipped with an Intel Xeon Gold 5215 processor running at 2.5GHz, 96GB of RAM, and an Nvidia Quadro RTX5000 graphics card with 16GB of memory. The operating system in use was Ubuntu 18.04 64-bit.

Fig. 7 Training and validation accuracy graph for training on NetSurfP–2.0 dataset

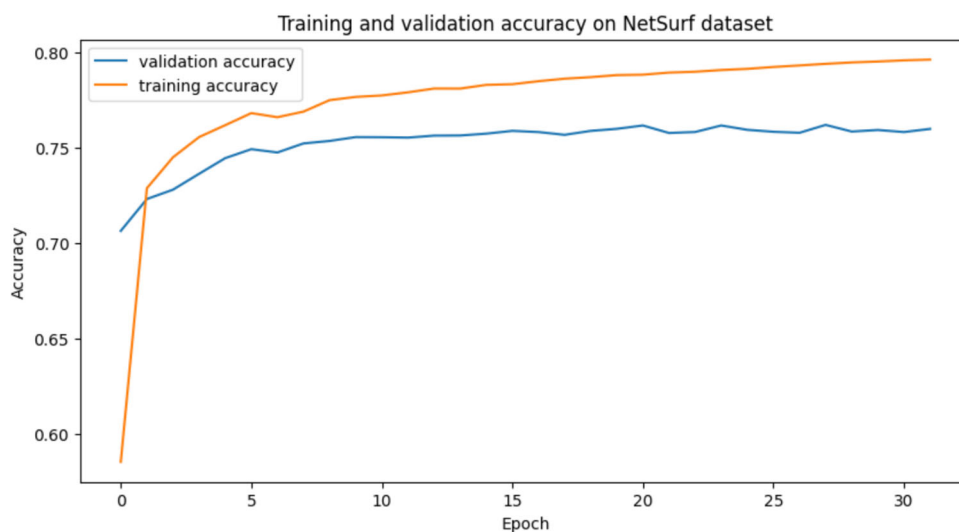


Table 2 Comparison of the accuracies of models trained on NetsurfP–2.0 dataset

Model	CASP12	NEW364	CB513	TS115
DeepProVec	49.7	53.3	48.9	54.4
DeepSeqVec	61	64.8	62.7	67.2
ESM-1b	66	71.3	70.2	73.4
ProtT5-XL-U50	68.9	74.5	74.6	77
ProtT5-XXL-U50	68.1	72.5	71.6	75.1
NetsurfP–3.0	66.4	72.9	72	75.7
NetsurfP–2.0 (profile)	70.3	73.9	72.3	75
AttSec	70.6	75.5	75.2	78.5
TransConv (ours)	71.3	75.9	76.1	79.1

Bold entries signify the best result among the other compared models

Results and discussions

In this study, we evaluate the performance of the proposed TransConv model in comparison to several SOTA models for protein secondary structure prediction. The TransConv model was trained separately on two distinct datasets and its performance was assessed across various test datasets.

Table 2 presents a comparative analysis of the average accuracy of various models trained on the NetSurfP–2.0 dataset. The results demonstrate that the TransConv model consistently outperforms existing methods across all test datasets. Specifically, TransConv achieves an average accuracy of 71.3% on CASP12, 75.9% on NEW364, 76.1% on CB513, and 79.1% on TS115. These significant improvements over other models highlight TransConv’s effectiveness in capturing both local and long-range dependencies in protein sequences.

Additionally, the performance of the TransConv model was evaluated on the ProteinNet dataset and compared with other established models. Table 3 summarizes these results. The TransConv model achieves superior performance across all the test datasets within the ProteinNet dataset, with average accuracies of 78.0% on TEST2018, 70.4% on SPOT-2016, 73.4% on SPOT-2016-HQ, 71.4% on SPOT-2018, and 72.6% on SPOT-2018-HQ. These results further under-

score the effectiveness of TransConv in accurately predicting protein secondary structure, surpassing the performance of existing methods.

The current SOTA model, AttSec [27], is a transformer-based model that uses the ProtT5-XL-U50 language model to generate input embeddings. It employs an embedding size of 1024 and has three encoder layers with eight attention heads each. AttSec obtains attention weights from the transformer encoder and applies 2D convolution to these weights to generate the final output for secondary structure classification. Our proposed TransConv model achieves higher accuracy than the AttSec model when trained on both the NetSurfP–2.0 dataset and the ProteinNet dataset. Figures 8 and 9 compare the accuracies achieved by the AttSec [27] and TransConv models.

For the evaluation of our models, we selected the one with the highest validation accuracy during training, rather than the model from the final epoch, to ensure better generalization on the test set. For the model trained on the NetsurfP–2.0 dataset, the 27th epoch produced the best validation accuracy, and this was the model used for final testing, with training continuing until the 31st epoch before early stopping was applied due to overfitting. Similarly, for the model trained on the ProteinNet dataset, the best validation accuracy was achieved at the 10th epoch, which was then used for testing, with training proceeding until the 21st epoch before early stopping. This approach allowed us to prevent overfitting and report the most accurate performance of the models on unseen data.

Tables 4 and 5 present the precision, recall, and F1-score of the TransConv model for each test dataset. These metrics are calculated separately for each of the eight classes corresponding to the 8-state secondary structure, and an average across the classes is then computed to derive the final values reported in these tables. This comprehensive evaluation provides insights into the model’s performance across different categories of secondary structure prediction, beyond simple accuracy.

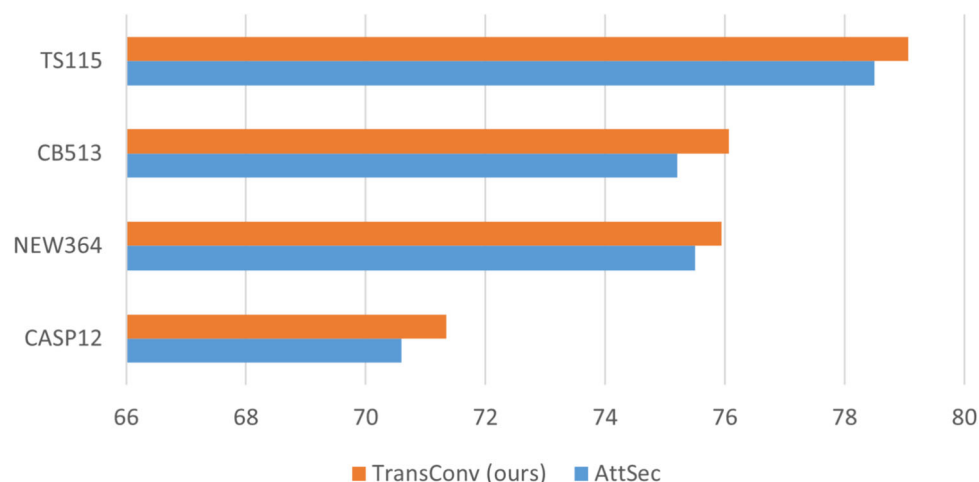
Furthermore, Table 6 presents precision, recall, and F1-score comparisons across models trained on the NetSurfP–2.0 dataset, with the results for most models sourced from pub-

Table 3 Comparison of the accuracies of models trained on ProteinNet dataset

Model	TEST2018	SPOT-2016	SPOT-2016-HQ	SPOT-2018	SPOT-2018-HQ
PSIPRED-Single	–	–	–	–	–
SPIDER3-Single	59.8	58.9	59.9	57.4	58.0
ProteinUnet	60.3	–	–	–	–
SPOT-1D-Single	62.2	61.4	61.6	60.1	60.0
SPOT-1D (profile)	75.4	69.3	71.7	67.4	70.5
AttSec [27]	77.4	70.4	73.0	71.2	72.3
TransConv (ours)	78.0	70.4	73.4	71.4	72.6

Bold entries signify the best result among the other compared models

Fig. 8 Accuracy of proposed model and AttSec model trained on NetSurf 2.0 DSSP8 dataset



lished studies or re-implementations. Data for DeepProtVec and ESM-1b on the NetSurfP-2.0 dataset were unavailable in existing literature, and re-implementation did not yield comparable results due to limited access to specific training parameters and preprocessing details. Therefore, results for these two models are not presented in the comparison table. From Table 6, the results show that our proposed TransConv model outperforms the existing methods across all three metrics. Specifically, TransConv achieves the highest precision (0.63), recall (0.53), and F1-score (0.56), indicating its superior performance in predicting protein secondary structures. In comparison, other models, such as ProtT5-XL-U50 and AttSec, exhibit lower values, highlighting the effectiveness of the convolution-infused transformer architecture in our approach.

Ablation study

An ablation study was conducted to examine the impact of input embedding size on the performance of the transformer

encoder model. In this study, we converted input protein sequences to embeddings using the pre-trained ProtT5-XL-U50 model, which produces embeddings of size 1024. To adapt these embeddings to our model's requirements, we applied principal component analysis (PCA) to reduce the embedding size to the desired dimensions.

In our experiments, we primarily utilized an embedding size of 128. However, it is essential to investigate the effects of different embedding sizes on model performance. Smaller embedding sizes yield a more compact model with fewer parameters, potentially reducing computational requirements and mitigating the risk of overfitting. In contrast, larger embedding sizes increase the model's capacity by adding more parameters, which can enhance the model's ability to capture complex patterns but also lead to longer training times and a higher risk of overfitting. Therefore, achieving an optimal balance between embedding size and model complexity is crucial for attaining high accuracy without excessively increasing computational demands.

The results of training the model with various input embedding sizes are presented in Table 7. As shown, the aver-

Fig. 9 Accuracy of proposed model and AttSec model trained on ProteinNet DSSP8 dataset

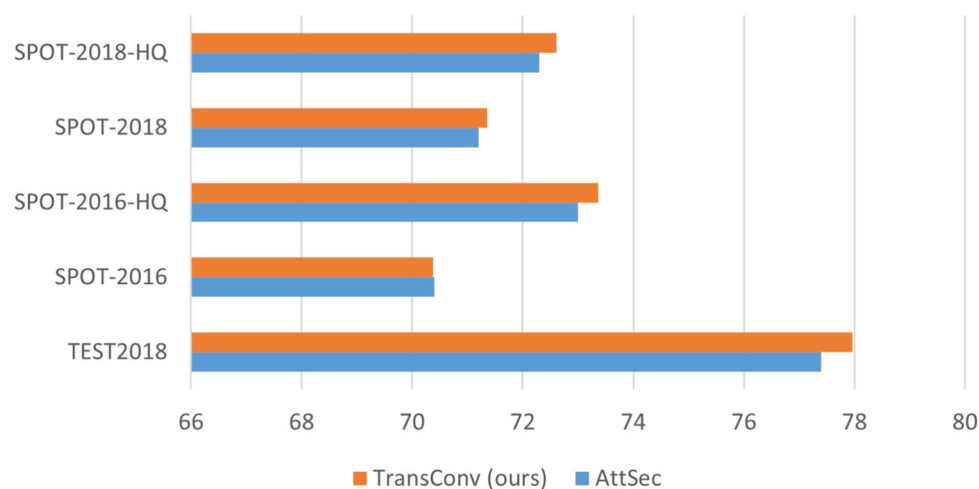


Table 4 Performance metrics for proposed model trained on NetsurfP–2.0 dataset

Test dataset	Precision	Recall	F1-score
CASP12	0.64	0.49	0.53
NEW364	0.57	0.48	0.50
CB513	0.65	0.57	0.60
TS115	0.67	0.59	0.61

Table 5 Performance metrics for proposed model trained on ProteinNet dataset

Test dataset	Precision	Recall	F1-score
TEST2018	0.57	0.51	0.53
SPOT-2016	0.59	0.38	0.39
SPOT-2016-HQ	0.52	0.42	0.43
SPOT-2018	0.47	0.38	0.39
SPOT-2018-HQ	0.53	0.41	0.43

Table 6 Comparison of precision, recall, and F1-score of models trained on NetsurfP–2.0 dataset

Test dataset	Precision	Recall	F1-score
DeepProtVec	–	–	–
DeepSeqVec	0.47	0.48	0.49
ESM-1b	–	–	–
ProtT5-XL-U50	0.53	0.49	0.51
ProtT5-XXL-U50	0.54	0.49	0.48
NetsurfP–3.0	0.57	0.48	0.50
NetsurfP–2.0 (profile)	0.59	0.47	0.48
AttSec	0.60	0.51	0.52
TransConv (Ours)	0.63	0.53	0.56

Table 7 Average test accuracy for different embedding sizes across the various test datasets when trained on NetSurfP–2.0 dataset

Embedding size	CASP12	NEW364	CB513	TS115
32	71.1	75.1	75.4	78.9
64	71.1	75.6	75.7	79.1
128	71.3	75.9	76.1	79.1
256	29.4	36.5	32.1	38.7

age accuracies improve as we increase the embedding size from 32 to 64 and then to 128. However, when the embedding size is further increased to 256, the model struggles with convergence and fails to reach an optimal solution. This difficulty in convergence for larger embedding sizes can be attributed to the increased number of trainable parameters, which complicates the optimization process. A potential solution to this issue could be adjusting the learning rate; for example, a smaller learning rate might facilitate better convergence for models with larger embeddings. It is important to note that this ablation study was conducted using a consistent learning rate schedule across all training sessions to ensure the comparability of results.

Conclusion

This paper introduces a novel hybrid model for protein secondary structure prediction, combining convolution operations with a transformer encoder to effectively capture both local patterns and long-range dependencies in protein sequences. The convolutional layers excel in identifying short-range interactions, while the attention mechanism of the transformer encoder models the broader context across amino acids, contributing to the overall accuracy of the predictions. Our model demonstrates promising results, outperforming several baseline approaches in terms of prediction accuracy. However, we acknowledge certain limitations, such as the risk of overfitting observed during training and the need for further validation across diverse datasets to ensure robustness and generalizability. Moreover, while the current model configuration shows improvement, there is potential for further optimization of hyperparameters using advanced meta-heuristic algorithms like genetic algorithms or Bayesian optimization. Future work can also explore expanding the model to predict 3-state secondary structures, enhancing its utility for broader applications in computational biology.

Author Contributions Sayantan Das, Subhayu Ghosh, and Nanda Dulal Jana contributed to the study conception and design. Coding implementation and analysis, and the writing of the first draft of the manuscript were done by Sayantan Das. Subhayu Ghosh participated in revising the paper and completing the final draft. Nanda Dulal Jana guided throughout the research and supervised every aspect of this work.

Data availability The source code of the proposed Trans-Conv is publicly available at: <https://github.com/sayantanDs/transconv>.

Declarations

Ethical approval Not applicable

Informed consent Not applicable

Competing interests The authors declare no competing interests.

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